



Proposed CEA Link — December 2026

Weeks 17-18 Swapped re-ordered by hand to cut fatigue — same duties, same days, same coverage; only which week sits next to which has changed

Prepared 22 September 2026 · re-rendered 25 September 2026 · built for the December 2026 timetable and assessed by the workspace's own rule modules (Marylbone Roster v24.22) · figures on this page are computed, not typed

PROPOSAL

Targeted Fatigue Redo

Weeks 17-18 Swapped re-ordered by hand — same duties, same days, same coverage

CODE **TF-24-EXT**
FINGERPRINT **8eef9a13**
BUILT FROM duty table — · seed — · 24 lines · 4 cover weeks

✓ **24** lines designed ✓ **35h 00m** a week — exactly the contract ✓ **0** hard-limit breaches ✓ **0** rests under 12h

⚠ **4** fatigue factors present — 3 as rostered (today: 5)

Five questions a manager asks first, in order. Every figure is computed from the cells; **page 2 says how to read each one**, with today's link and the rest of the folder marked on a scale.

CAN IT BE RUN?

Yes

the hard limits

0 rests under 12h · longest run 9 of 13 · contract exact

DOES IT MEET THE DECEMBER SHAPE?

0 of 4

headcount rules met

four to open, three to close (four on Saturday), five at 22:00, 14 and 10 at the weekend · today meets 0 of 4

HOW TIRING IS IT?

4

fatigue factors present, of 25

advisory — present means worth a look, not a breach · today 5 · fewest in the folder 0

DOES IT FOLLOW THE TRAINS?

34.4

weekday demand fit — lower is closer

Saturday 20.7 · Sunday 62.4 · today 44.7 · best in the folder 30.7

IS IT FAMILIAR?

13 of 19

shift times people work today

6 new, listed on page 5 · 4 of 20 weeks are one turn (today 7 of 16)

5 in 24

full weekends off

21% of the rotation, against 20% today (4 in 20) · most in the folder 25%

4 cover weeks

lines 1, 7, 12, 18

gaps of 7, 6, 5, 6 around the wheel — close to even, one line out; unchanged from the design this re-orders

4h 10m a week

typical week-to-week move

how far the working day shifts each week — today's link moves 4h 0m a week

What this is

A proposal for the CEA link once the December 2026 timetable arrives, for a team of 24 rather than 20. The brief was to keep the **feel** of the link people work today — a week is one turn, rest days come in pairs, four whole cover weeks spread round the rotation, and the same 06:20 / 08:00 / 14:00 / 15:15 clock times — while meeting the staffing shape agreed for December 2026 (four to open every day, three through to the close, four on a Saturday, five still on at 22:00, fourteen on a Saturday, ten on a Sunday) and every rule the link is assessed against: Chiltern's 13-day limit, twelve hours between duties, the exact 35-hour contracted week, and the ORR's good-practice fatigue factors.

It was not drawn by this app. It arrived as a Word table — a proposal from outside the workspace — and was read straight into the app's own shape, then judged by the same modules that judge a design the workspace generated itself: runDesignChecks, weeklyHours, assessHardLimits, assessFatigue, scoreOrder and calcHourlyCoverage. There is no search behind it and nothing to reproduce; what can be checked is every figure on these pages, each computed from the cells on page 8. Paste those cells into Links → Import and the workspace restates all of it.

What it is not. Not a recommendation and not a finished roster: nothing here has been through the roster office or a rep. The 24-line length, the four cover weeks and the December 2026 headcounts are owner-relayed figures with no document behind them (evidence class C), and the 13-day limit's policy citation is still outstanding — pages 7 and 8 say so where it applies. It clears every rule the tool can check, which is the floor for being worth discussing.

HOW TO READ THE NUMBERS

What each figure means, and what good looks like

Every figure on this sheet is computed from the cells. On each scale: this sheet in navy, today's 20-line link as a ring, the best and worst of the other 21 sheets in the folder as gold and grey ticks.

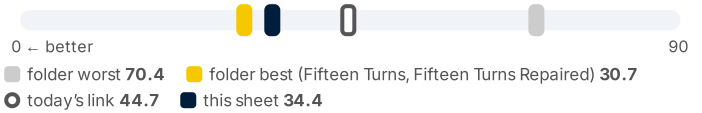
Three things to hold. Lower is better on the fit and the fatigue count; higher is better on rest, weekends, one-turn weeks and familiarity. "Present" is not "failed" — the ORR list reports what is there. Today's link is the yardstick, not the target: it meets none of the December 2026 headcounts.

DOES IT FOLLOW THE TRAINS?

Demand fit — weekday

How closely the people on duty follow the trains through an average weekday. For each hour, its share of the day's staff-minutes is compared with its share of the timetable's traffic; the gaps are squared and added up. 0 would be a perfect match; a score near 25 means each hour is about one percentage point off its share, near 50 about one and a half.

What it does not tell you. It compares shares, not headcount — adding people evenly to every hour changes nothing — and the station rosters posts that need someone whether the hour carries 23 trains or 5, so a flatter day can be right and still score worse.



Saturday **20.7** (today 45.7, folder best 8.1) · Sunday **62.4** (today 65.9, best 20.9). Each day type has its own curve and window, so a Saturday is only compared with a Saturday.

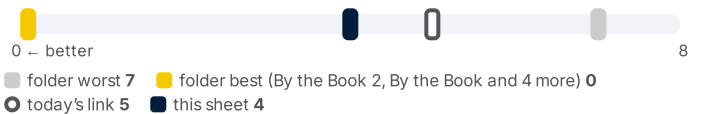
On page 6.

HOW TIRING IS IT?

Fatigue factors present

How many of the 25 roster patterns on the Office of Rail and Road's good-practice list appear anywhere in the rotation — long runs of earlys, short gaps between duties, start times that jump — counted by the workspace's own module.

What it does not tell you. A factor present is worth a look, never a breach: the ORR says the list is guidance, not limits. Zero does not mean approved and one does not mean unsafe; page 8 names which, and how big.



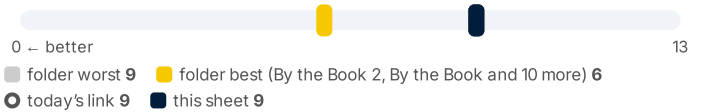
On page 8.

CAN IT BE RUN?

Longest run of worked days

The most consecutive days anyone works in the worst case — a cover week's four duties placed as badly as they can be. Chiltern's limit is 13; the workspace's own target is 6.

What it does not tell you. The roster clerk decides where cover-week duties fall, so the as-rostered figure can be lower; page 8 shows both readings.



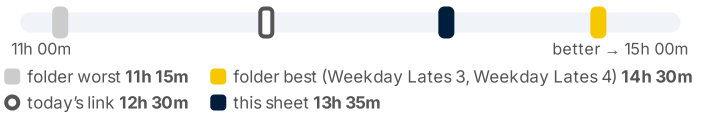
On page 7.

CAN IT BE RUN?

Tightest rest between duties

The shortest gap anywhere between one duty ending and the next beginning — Saturday into Sunday and the last line into the first included. Twelve hours is the floor; a design under it cannot be run.

What it does not tell you. It is one gap. A rotation with many 12h05 rests and one with none read the same here; the count under 12h is in the chips on page 1.



On page 7.

WHAT IS IT LIKE TO WORK?

Full weekends off

The share of weeks with both Saturday and Sunday off. Today's link gives 4 weekends in 20; a 24-line link with six is the same one weekend in four.

What it does not tell you. A weekend off beside a 42-hour week is still a weekend off; the hours by week are on the grid, page 4.



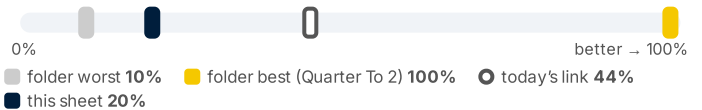
On page 5.

WHAT IS IT LIKE TO WORK?

Working weeks on one turn

The share of working weeks where a person keeps the same clock time all week. A week that mixes an early and a late is what people mean by "all over the place".

What it does not tell you. One turn all week can still be a bad week if the turn is long; the week-to-week move and the duty lengths are on pages 5 and 8.



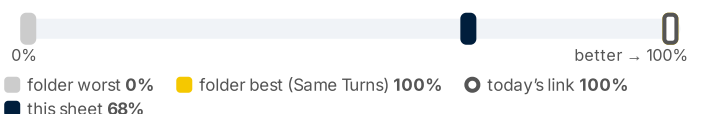
On page 5.

IS IT FAMILIAR?

Shift times people work today

The share of this sheet's clock times that somebody on the 20-line link already works. It is the folder's proxy for how much there is to learn and to accept.

What it does not tell you. Familiar is not the same as popular — a time everyone works and nobody likes still counts — and a new time in one week is not the same as one in ten. The duty table on page 5 lists the new ones.



On page 5.

BEFORE THE FIGURES

The decision this sheet supports

What is being decided, which criteria are hard and which are the room's, how much each figure can bear, and what no page here can tell you. Read this before the tiles.

- 1 What this is

4 The 24-line grid, with the words explained

7 Hard limits and the December 2026 figures

10 The grid, ready to paste into the app
- 2 How to read the numbers

5 A week compared with today

8 The ORR fatigue factors
- 3 The decision it supports (this page)

6 People on duty, hour by hour

9 Where it came from

The question

Should the link for the December 2026 timetable be the hand-built Weekday Lates line, in this revision, accepting 12 on a Saturday against the agreed 14 and 0 of the four December 2026 headcount rules met, for a week shape its authors chose?

Targeted Fatigue Redo is read two ways: against Quarter To (QT-24-Q34) and By the Book (BB-24-D7), the two searched answers; and within its own line, against Weeks 17-18 Swapped, the parent whose changed cells are outlined on the next page.

The criteria, in three tiers

HARD — A DESIGN MEETS THESE OR CANNOT BE RUN

- ✓ Twelve hours between duties — met, tightest 13h 35m
- ✓ Chiltern's 13-day limit — longest run 9 days
- ✓ The 35-hour contract, exactly — 35h 00m

SOFT — THE ROOM WEIGHS THESE AGAINST EACH OTHER

- December 2026 headcounts met: **0 of 4** (today 0 of 4)
- Fatigue factors present: **4** worst case, 3 as rostered (today 5)
- Demand fit weekday · Sat · Sun: **34.4** · 20.7 · 62.4 (today 44.7 · 45.7 · 65.9; lower is closer)
- Times worked today: **13 of 19** · full weekends off: **5** in 24 (today 4 in 20)

PREFERENCE — STAFF'S TO STATE, NOT THE TOOL'S

- Early against late: earlies 7h 15m–8h 30m, lates 7h 30m–9h 10m
- The 6 new times: 07:00-15:30, 08:00-16:00, 08:30-17:00, 10:30-19:00, 12:00-20:30, 16:25-23:55
- The longest duty, 9h 10m, and who holds it
- Cover weeks at lines 1, 7, 12, 18

How much to trust each figure

FIGURE ON THESE PAGES	WHERE IT COMES FROM	WEIGHT
0 rests under 12h · 35h 00m contract · 12/9 at the weekend	Computed from the cells; re-computed by the workspace on import	Firm
Longest run 9 days, inside 13	Computed; the 13-day limit's policy citation is still outstanding	Firm figure , unconfirmed threshold
Four at the open, three to close, 14 and 10 at the weekend	Owner-relayed December 2026 headcounts with no document behind them (class C)	Provisional
4 fatigue factors present	The ORR's 2021 list, which it states are not limits; 4 factors still "definition to confirm"	Advisory only — never pass or fail
Demand fit 34.4 · 20.7 · 62.4	Measured timetable, one share-based measure on one December 2026 timetable curve	Indicative
13 of 19 times are today's	Computed against the live 20-line link	Firm , but a proxy for acceptability
6 cells changed against Weeks 17-18 Swapped	Computed cell by cell; outlined on the next page	Firm

What no page here can tell you

- Whether the people who would work them accept the times nobody works today (07:00-15:30, 08:00-16:00, 08:30-17:00, 10:30-19:00, 12:00-20:30, 16:25-23:55) — familiarity is measured as "times worked today", which is a proxy for acceptability, not acceptability.
- Whether the roster clerk splits a cover week day-on-day-off. It is the one thing that moves the 48-hour-break figure (12 as a block, 15 split), and the link does not decide it.
- How leave and sickness cover land on a 24-line link with four cover weeks (lines 1, 7, 12, 18); nothing here models absence.
- Whether the December 2026 timetable demand curve holds on Wembley event days — the Saturday weighting approximates them, and the curve is a measured timetable, not a footfall count.

Before it is frozen

Lines 2, 6, 9, 16, 17 hold the times nobody works today; the people who would work those weeks read them first and say in one line each what they would change. The roster office answers the cover-week question. Someone confirms the source of the 13-day limit. Only then does the factor count on page 8 mean anything, and only then is this sheet more than an aid to a conversation.

THE ROTATION

The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

LINE	SUN	MON	TUE	WED	THU	FRI	SAT	MON-SAT hours	ALL WEEK hours	DAYS WORKED Mon-Sat	TURNS Mon-Fri
1	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
2	14:30 23:25	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	RD	RD	30h 00m	38h 55m	4	1
3	07:15 15:45	RD	08:30 16:30	06:20 14:20	06:20 14:20	06:20 14:20	06:20 14:20	40h 00m	48h 30m	5	2
4	RD	RD	06:20 14:20	08:30 16:30	RD	13:30 22:00	14:45 23:55	33h 40m	33h 40m	4	3
5	14:30 23:25	RD	14:00 22:30	14:00 22:30	14:00 22:30	16:25 23:55	RD	33h 00m	41h 55m	4	2
6	RD	06:20 14:20	07:00 15:30	07:00 15:30	07:00 15:30	08:30 17:00	RD	42h 00m	42h 00m	5	3
7	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
8	07:15 15:45	06:20 13:45	RD	RD	13:30 22:00	14:00 22:30	13:30 21:00	31h 55m	40h 25m	4	3
9	RD	12:00 20:30	12:00 20:30	12:00 20:30	12:00 20:30	12:00 20:30	RD	42h 30m	42h 30m	5	1
10	RD	14:00 22:30	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	14:45 23:55	47h 40m	47h 40m	6	2
11	RD	07:00 15:30	07:00 15:30	RD	RD	06:20 13:35	06:20 14:50	32h 45m	32h 45m	4	2
12	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
13	RD	RD	14:00 22:30	13:30 22:00	14:00 22:30	14:00 22:30	14:30 22:00	41h 30m	41h 30m	5	2
14	RD	16:25 23:55	16:25 23:55	RD	RD	07:00 15:30	06:20 14:50	32h 00m	32h 00m	4	2
15	07:15 15:45	06:20 13:45	06:20 13:35	06:20 13:35	06:20 13:35	07:00 15:30	RD	37h 40m	46h 10m	5	3
16	RD	08:30 17:00	08:30 17:00	08:30 17:00	RD	RD	RD	25h 30m	25h 30m	3	1
17	10:30 19:00	16:25 23:55	RD	RD	06:20 14:20	06:20 14:20	06:20 14:50	32h 00m	40h 30m	4	2
18	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
19	14:30 23:25	RD	06:20 13:45	06:20 13:45	06:20 13:45	06:20 13:45	08:00 16:00	37h 40m	46h 35m	5	1
20	08:30 16:30	06:20 13:45	RD	RD	13:30 22:00	16:25 23:55	13:30 22:00	31h 55m	39h 55m	4	3
21	13:30 22:00	13:30 22:00	RD	13:30 22:00	13:30 22:00	13:00 21:00	RD	33h 30m	42h 00m	4	2
22	RD	14:00 22:30	13:30 22:00	13:30 22:00	13:30 22:00	RD	RD	34h 00m	34h 00m	4	2
23	RD	RD	RD	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	30h 00m	30h 00m	4	1
24	RD	RD	06:20 13:45	06:20 13:45	07:00 15:30	RD	06:20 13:45	30h 45m	30h 45m	4	2
Cover	9 SP 4	13 SP 4	15 SP 4	15 SP 4	16 SP 4	16 SP 4	12 SP 4	35h 00m	38h 13m	4.35	4 one turn
Minutes	4,635	6,225	7,205	7,205	7,780	7,690	5,895	42,000	46,635		

Early turns Late turns Cover (spare) week — four duties of seven, any turn Rest day

Changed against *Weeks 17-18 Swapped (WS-24-EXT)* — 6 cells on 6 lines

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines · Turns: distinct clock times Monday to Friday, **bold** where the week is one turn (one clock time Mon-Fri and one early-or-late family across every worked day — the page 1 figure) · Minutes: duty minutes per day, and Mon-Sat is the contract (20 × 35h = 42,000)

The words on this page

Link — the whole rota: 24 weeks laid out as 24 lines. Everyone works line 1, then line 2, and so on round the wheel; the link needs 24 people.

Turn — a shift time, such as 06:20-13:45. An **early** starts before 09:00; a **late** starts after.

Cover (spare) week — a week with no fixed times. You work four duties of the seven days, placed by the roster clerk to cover leave and sickness.

The contract — 35 hours a week, averaged over the 24 lines; a cover week counts as a full contracted week.

Line — one week of the pattern, Sunday to Saturday. You move down a line each week, and line 24 goes back to line 1.

Rest day (RD) — a day off inside the contracted week. Two together make a proper break; one on its own does not count as a 48-hour break.

Sunday — not part of the contract. A Sunday duty is rest-day working, paid on top of the week, which is why the hours are shown Mon-Sat and all week.

One-turn week — the same clock time all week (shown bold in the Turns column); the easiest kind of week to live around.

Same feel, more people

What this design keeps of today, and where the December 2026 headcounts differ

Kept — the shape of a week

MEASURE	TODAY (20)	PROPOSED (24)
Working weeks that are one turn (one clock time Mon–Fri)	7 of 16	4 of 20
Weeks mixing early and late turns	7	6
Rest-day breaks that are two days or more	13 of 17	15 of 25
Single rest days between duties (not a 48h break)	4	10
Days worked in a week (lines × days)	1×3, 3×4, 11×5, 1×6	1×3, 6×4, 9×5, 4×6
Cover (spare) weeks (whole weeks)	lines 1, 7, 12, 17	lines 1, 7, 12, 18
Distinct shift times (all but 6 on today's roster)	18	19
Mon–Sat hours, average week (the contract)	35h 00m	35h 00m
Sunday duties (not contracted; half the working lines)	8 of 16 lines	9 of 20 lines

Changed — the December 2026 headcount

PEOPLE ON DUTY	TODAY	PROPOSED	WHY
Monday to Friday	11–12	13–16	the contract: 20 working lines × 35h has to be worked somewhere
Saturday	10	12	owner's figure for December 2026, leaning late for events
Sunday	8	9	owner's figure for December 2026
Opening at 06:20 (07:15 Sunday) wk · Sat · Sun	4 · 4 · 3	4 · 5 · 3	four to open, every day — not met: Saturday 5, Sunday 3
Through to the close (23:55; 23:25 Sunday) wk · Sat · Sun	2 · 2 · 3	3–4 · 3 · 3	three to close, four on a Saturday — not met: Fri 4, Saturday 3
Still on duty at 22:00 wk · Sat · Sun	4 · 2 · 3	4–6 · 3 · 3	five from 22:00, Monday to Saturday — not met: Wed 4, Fri 6, Saturday 3

The duty table, in today's times

TURN	WK	SAT	SUN	WK	SAT	SUN
06:20–13:35 7h 15m · Early	1		1			
06:20–13:45 7h 25m · Early	1		3 +2	1 +1		
06:20–14:00 7h 40m · Early		4		4		
06:20–14:20 8h 00m · Early	2		2	1 +1		
06:20–14:50 8h 30m · Early		3		3		
07:00–15:30 8h 30m · Early			2 +2			
07:15–15:45 8h 30m · Early			3			3
08:00–16:00 8h 00m · Early				1 +1		
08:00–16:30 8h 30m · Early	2		2			
08:30–16:30 8h 00m · Early			1	1 +1		1
08:30–17:00 8h 30m · Early				1 +1		
10:30–19:00 8h 30m · Early						1 +1
11:00–19:30 8h 30m · Late	4		4			
12:00–20:00 8h 00m · Late		4		4		
12:00–20:30 8h 30m · Late			1 +1			
13:00–21:00 8h 00m · Late			1	1 +1		-1
13:30–21:00 7h 30m · Late		1		1		
13:30–22:00 8h 30m · Late	1		4 +3	1 +1	1 +1	
14:00–22:30 8h 30m · Late	2		2			
14:30–22:00 7h 30m · Late		2		1 -1		
14:30–23:25 8h 55m · Late			3			3
14:45–23:55 9h 10m · Late		2		2		
15:15–23:55 8h 40m · Late	2		2			
16:25–23:55 7h 30m · Late			4 +4	1 +1		

Busiest weekday · Saturday · Sunday. The first three columns are what the 20-line link does now; the last three are the proposal. Green cells grew, amber shrank, a struck-through row is a time the proposal does not use.

Why the duties land where they do: Monday to Saturday totals exactly 42,000 minutes, which is 20 × 35h to the minute, so the contract is paid. But unlike a table built day-uniform, **the weekdays here are not equal** — Mon 6,225 · Tue 7,205 · Wed 7,205 · Thu 7,780 · Fri 7,690, and Saturday 5,895. Monday is the lightest day by some margin and Thursday the heaviest; that is a property of the design rather than a fault, but it is the first thing to put to the roster office, because it decides how much cover each day really has.

People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's traffic.

How to read it: each blue cell is the number of people on duty in that hour (darker blue = more people). The orange row is the timetable — how much of the day's trains that hour carries (darker orange = busier). Read the columns top to bottom: cover should be thickest where the orange is darkest. The **fit** figure at the right end of each row scores that match for the whole day.

Monday to Friday window 06:20–23:55 · one row per distinct weekday

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY MON · WED		4	4	6	6	6	7	7	7	7	7	7	5	5	5	4	4	4	2	55.8
TODAY TUE · THU · FRI		4	4	6	6	6	7	7	8	8	8	8	6	6	6	5	5	4	2	50
PROPOSED MON		4	5	6	6	6	6	7	8	7	6	8	7	7	7	7	6	5	3	28.4
PROPOSED TUE		4	6	8	8	8	8	9	10	9	8	9	7	7	7	7	6	5	3	43.6
PROPOSED WED		4	5	7	7	7	7	8	11	9	8	10	8	8	8	8	7	4	3	42.3
PROPOSED THU		4	6	6	6	6	6	7	11	11	9	10	10	10	10	10	9	5	3	48.8
PROPOSED FRI		4	6	7	7	7	7	8	10	10	8	10	9	9	9	9	7	6	4	31
DEC 2026 TRAFFIC	5	69	107	127	122	95	85	75	84	79	87	105	140	138	107	90	73	75	66	

Saturday window 06:20–23:55

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY		4	4	4	4	4	4	5	6	9	6	6	6	6	6	5	4	2	2	45.7
PROPOSED		5	5	6	6	6	6	6	8	10	6	6	6	6	6	6	5	3	3	20.7
DEC 2026 TRAFFIC	6	24	58	75	81	92	88	77	64	68	72	84	82	67	64	79	78	62	41	

Sunday window 07:15–23:25

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY			3	4	4	4	4	4	5	8	8	5	4	4	4	4	3	3	3	65.9
PROPOSED			3	4	4	5	5	5	6	9	9	6	5	5	4	4	4	3	3	62.4
DEC 2026 TRAFFIC			15	45	65	71	74	71	61	55	64	69	78	73	75	52	71	65	50	

Spare cover is not in these figures — a cover week carries no times, so the rows are a floor. **Fit** is one number for how closely the people on duty follow the trains through the day: 0 would be a perfect match, lower is better, and today's link scores what it scores. (For the record: the squared gap between each hour's share of the day's traffic and its share of the cover, inside the day's window, measured on duty minutes — so an hour a duty only touches counts in proportion; the cells above count heads.) It is the measure the searched tables were chosen on, and the fit of the average weekday is the *Wk fit* on page 9. Sunday: 5 December 2026 timetable movements fall after the 23:25 finish — the standing question on whether Sunday's window moves; the window is stored per design, so this proposal can be rebuilt to either answer.

Reading it. The proposal raises the whole day rather than reshaping it, which is what the December 2026 timetable brief asked for — the station rosters posts (ticket office, gateline, passenger assist) plus break cover, and a post needs someone on it whether the hour carries 23 trains or 5. The weekday's two peaks are a duty length apart, so cover stays fairly flat between them by arithmetic, not neglect.

The checks sheet

Hard limits first — met or not. Then the ORR good-practice fatigue factors, which report what is present and never pass or fail a design.

Hard limits — a design either meets these or cannot be run

- ✓ **More than 13 consecutive days worked** — longest possible run **9** days (today: 9)
Longest possible run is 9 days, within the 13 configured here from Chiltern practice (origin: the legacy Hidden standard). A spare week is 4 duties of 7 and the roster clerk places them, so this is the worst case the link allows. Worked as a BLOCK of four rather than split day-on-day-off, the answer is the same — **9** days — so this row does not depend on how a cover week is placed.
Basis: Chiltern roster policy, citation outstanding — legacy Hidden 13-in-14 standard. Configured from Chiltern practice; the policy citation is outstanding, so this is stated as the app states it.
- ✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today: 0)
Tightest anywhere: **13h 35m** — **line 20 Fri 16:25-23:55 into Sat 13:30-22:00** (today's tightest: 12h 30m). The generator refuses a design it cannot repair to this; this design was checked against it, not generated.
- ✓ **The contracted week, exactly** — **35h 00m** average Mon-Sat over the 24 lines, cover weeks counted as contracted weeks
700h of duty a week across 20 working lines. Sundays (77.25h) sit on top as RDW, as they do today. Individual weeks range 25h 30m–47h 40m; only the average is the contract.

December 2026 timetable design figures — the staffing shape agreed for the new timetable

RULE	PROPOSAL	
Four on at the open, every day	× 4 weekday · 5 Saturday · 3 Sunday	
Three through to the close; four on a Saturday	× 3–4 weekday · 3 Saturday · 3 Sunday	
Five still on duty at 22:00	× 4–6 weekday · 3 Saturday	a 22:00 finish is not "on at 22:00"
Fourteen on a Saturday, ten on a Sunday	× 12 · 9	
Cover weeks, evenly spread	× 4 weeks at lines 1, 7, 12, 18 — gaps 7, 6, 5, 6	the December 2026 staffing shape asks for four, evenly spread — the count is right; the spread is not
About 4.2 days a week worked, Mon-Sat	✓ 4.35 over the 20 working lines	
No :05 or :10 times except the open and close	✓ none	
Late turns slightly shorter than most earlies	× not met — lates 7h30–9h10, earlies 7h15–8h30	the longest late is longer than the longest early

To settle before it is frozen

One of the two targets was hit outright; the other depends on how a cover week is worked. The document sets out to clear two flags. The first is cleared on any reading: the worst seven-day total falls from **60.6 hours to 54.7**, under the 55-hour row, and the document's own estimate of "about 54h40" is right. The second is the interesting one. It reports the run "dropping from about 15 duties to 9", and there are three different measures in that sentence. **9** is the longest run of consecutive worked DAYS — correct, and inside Chiltern's thirteen. **FF11** counts shifts between **48-hour** breaks, and a single rest day is not one. On that measure this design is at **15 in the worst case**, down from 16 and over the threshold of 13 — but **12 as rostered**, under it. The difference is entirely the cover week: the 15 needs one split day-on-day-off so that it supplies no break and joins the blocks either side, and a cover week worked as a block always supplies one. **So whether this design clears FF11 is a question for the roster office, not for the link**, and page 8 prints both. **What the re-ordering cost.** Factors present fall from five to four in the worst case (three as rostered), but not uniformly: FF19 rises from **7 to 8**, full weekends off fall from **6 to 5**, and single-turn weeks fall from **6 of 20 to 4**, so more weeks now ask somebody to hold more than one start time. **Every hard gate stays clear** — 35h to the minute, no rest under 12 hours, no hard-limit breach, longest run nine days inside the thirteen on either reading. **The unfinished question is the same one the design it re-orders left open.** None of what remains needs a duty to change: FF11, FF15, FF8b and FF19 are all properties of which week sits next to which. On this exact duty table, the same shape work done by the workspace's own search takes the factors to **one** and the run to **six or seven** with the coverage curve untouched — the three searched proposals opposite are that result. This one is listed as supplied, deliberately, so the hand-made baseline stays on the record.

ORR fatigue factors

Good practice guidelines — Fatigue Factors, p3 (December 2021). Δ present — the pattern is in this design and worth a look, not a breach · $\checkmark$ clear — it is not · $\bullet$ standing — true of the station itself, not of any design · — does not apply here. This is the list the link is assessed against.

The Office of Rail and Road lists 25 roster patterns that tend to tire people — long runs of earlys, short gaps between duties, start times that jump about. For each one this table asks whether the pattern is in the rotation, today and proposed, and how big it is. The list is guidance: a factor present is a question for the room, never a pass or a fail, and a design showing nothing is not thereby approved.

CODE	FACTOR	TODAY'S LINK	PROPOSED
FF1	Night shift covering 00:00–05:00 Time of day	--	--
FF2	Early shift starting 05:00–07:00 Time of day	$\bullet$ 24	$\bullet$ 25
FF3	Very early shift starting before 05:00 Time of day	$\checkmark$ 0	$\checkmark$ 0
FF4	Very early shift (before 05:00) over 8h Duty length	--	--
FF5	Day shift over 12h Duty length	$\checkmark$ 0	$\checkmark$ 0
FF7	Early shift over 10h Duty length	$\checkmark$ 0	$\checkmark$ 0
FF8b	Fewer than 2 days rest after a block of early starts Recovery time	Δ 3	Δ 5
FF11	More than 13 consecutive shifts without a 48h break Recovery time	$\checkmark$ 12 $\checkmark$ 9 as rostered	Δ 15 $\checkmark$ 12 as rostered
FF10	More than 4 consecutive 12h day shifts Cumulative	$\checkmark$ 0	$\checkmark$ 0
FF15	More than 4 consecutive early shifts in a rotating pattern Cumulative	Δ 6	Δ 7
MRSF	More than 12 consecutive day shifts Cumulative	$\checkmark$ 9	$\checkmark$ 9
MRSF	More than 55 hours worked in any 7-day period Cumulative	Δ 60.8	$\checkmark$ 54.7
MRSF	More than 7 consecutive 8h shifts (definition to confirm) Cumulative	Δ 9	$\checkmark$ 7
MRSF	More than 6 consecutive night or early shifts in a permanent pattern Cumulative	--	--
FF17	Backward rotating pattern (definition to confirm) Circadian	$\checkmark$ 10 backward / 12 forward	$\checkmark$ 12 backward / 17 forward
FF18	Rotating pattern of about a week (definition to confirm) Circadian	$\bullet$ 20-line rotation · typically 4h 0m a week	$\bullet$ 24-line rotation · typically 4h 10m a week
FF19	Successive start times varying by more than 2 hours (definition to confirm) Circadian	Δ 6	Δ 8
FF6	Night shift over 10h Night shifts	--	--
FF8	Less than 2 days rest after consecutive nights Night shifts	--	--
FF9	Less than 14h rest in any 24h for night shifts Night shifts	--	--
FF12	Only one day rest after night shifts Night shifts	--	--
FF14	More than 4 consecutive nights in a rotating pattern Night shifts	--	--
FF16	More than 3 consecutive night shifts over 8h Night shifts	--	--
FF20	First night shift Night shifts	--	--
FF13	Less than 12h rest in any 24h Intervals between duties	$\checkmark$ 0	$\checkmark$ 0

Present: 4 in the worst case, 3 as rostered (today 5) · standing: 2 · FF2 fires on every 06:20 duty, so it is a property of the station's opening time rather than of any design. **Two readings of a cover week, on the one row they move.** A cover week is worked four days of seven and the link does not say which four, so a run figure is a range. The headline number is the ceiling — the four split day-on-day-off, which supplies no 48-hour break and joins the blocks either side. It is reachable: of the 35 ways to place four duties in seven days, the 10 that leave no two rest days together produce exactly it. Worked as a BLOCK, which is what a cover week looks like on the roster, the three rest days fall together and the week always supplies a break — that is the *as rostered* figure beneath it. Checked on every row: FF11 is the only one where the two differ. Which reading applies is a question for the roster office, not for this sheet. This sheet is an aid to a conversation, not a fatigue risk assessment.

METHOD

Where it came from

Checkable rather than reproducible: there is no search behind it, and every figure is computed from the cells

Lineage. Supplied as a Word table and assessed here — not produced by the proposal search. It is a pure RE-ORDERING of “Weeks 17-18 Swapped” (WS-24-EXT · 0bebb675): identical duty table, identical per-day duty multiset and an identical hour-by-hour coverage curve, with only the order of the weeks changed. Compare with the searched proposals “Weekday Lates 2/3/4”, which are the same kind of work done by the annealer.

In plain terms. Nobody searched for this design. It was typed in from the table exactly as supplied and the app's own rule modules measured it; nothing below was chosen by a computer. Everything on these pages can be re-checked by pasting page 10 into the Links workspace.

Where this design came from

1 • It was re-ordered by hand, not searched. This rotation arrived as a Word table describing a series of named, deliberate moves against *Weeks 17-18 Swapped*— a same-day duty swap at a time, each one argued in the document. It was read straight into the app's own shape and judged; no grid was annealed, so there is no search to reproduce. What can be reproduced is every figure on these pages.

2 • The document's central claim was tested first, and it holds. It states that every duty moved stays on the SAME weekday, so the number and type of shifts available on each day are preserved. Checked against *Weeks 17-18 Swapped*: each day's multiset of duties is **identical**, and so is the hour-by-hour coverage curve — every quarter-hour of the whole week. Nothing was invented, nothing re-timed, and no day lost or gained a person.

3 • The reading was checksummed before anything was built on it. A grid read from a document can carry a silent transcription error, so the duty minutes were totalled against the contract first: Monday to Saturday comes to **42,000** minutes against 20 working lines × 35h = 42,000, exactly. Every one of the document's own 24 stated weekly totals also matches its cells to the minute.

4 • The judge is the workspace's own — runDesignChecks, weeklyHours, assessHardLimits, assessFatigue, scoreOrder and calcHourlyCoverage, with nothing typed by hand and nothing softened. Where the design misses one of the December 2026 timetable rules the table opposite says so rather than omitting the row.

Alternatives measured

DESIGN	RUN	FF PRESENT	WKENDS	ONE- TURN	STEP	WK FIT	SCORE
TF-24-EXT · 8eef9a13 — Targeted Fatigue Redo (this proposal)	9	4	5	4/20	4h 10m	34.4	—
Same Turns · d15e1b74 — the searched proposal	6	1	6	18/20	2h 33m	46.3	—
By the Book · 0f14abce — the searched proposal	6	0	6	12/20	2h 52m	30.8	—
WS-24-EXT — Weeks 17-18 Swapped, the design this re-orders · 0bebb675	9	5	6	6/20	4h 17m	34.4	—
WL-24-EXT — Weekday Lates, as supplied · a52ec588	9	5	6	7/20	4h 43m	38.3	—
WL4-24-F7 — Weekday Lates 4, searched fatigue-first · f0d403d6	7	1	6	10/20	2h 34m	34.2	—
FT-24-R21 — Fifteen Turns Repaired, searched · b76bf9e1	6	1	6	16/20	3h 27m	30.7	—
Workspace default · 29ec9810 (Dec 2026 table, generated)	7	5	5	0/20	4h 16m	30.8	—
Today's 20-line link (for scale)	9	5	4	7/16	4h 0m	44.7	—

How to read this table: nothing here was picked by a search — this design was supplied or derived, and the rows are the comparisons it is being judged against. *Wk fit* is how evenly the WEEKDAY cover follows the December 2026 timetable traffic curve (lower is better; today's link scores what it scores), so it says nothing about Saturday or Sunday. **Score** is the search's own feel objective and is blank here, because no search produced this design — a blank is not a bad score. The workspace default is the app's own December 2026 duty table, generated and reordered with every switch on.

LOAD IT

The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

Links → Import, then paste the whole block below. Line number, then Sunday to Saturday; SP is a cover week. The workspace re-runs every check on these pages from the pasted cells, so nothing here has to be taken on trust — and the two designers can edit it there like any other design.

The same rotation is also supplied beside this PDF as Targeted-Fatigue-Redo-TF-24-EXT-import.txt (tab-separated, pastes directly) and Targeted-Fatigue-Redo-TF-24-EXT.json (the app's own format).

1	SP	SP	SP	SP	SP	SP	SP
2	14:30-23:25	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	RD	RD
3	07:15-15:45	RD	08:30-16:30	06:20-14:20	06:20-14:20	06:20-14:20	06:20-14:20
4	RD	RD	06:20-14:20	08:30-16:30	RD	13:30-22:00	14:45-23:55
5	14:30-23:25	RD	14:00-22:30	14:00-22:30	14:00-22:30	16:25-23:55	RD
6	RD	06:20-14:20	07:00-15:30	07:00-15:30	07:00-15:30	08:30-17:00	RD
7	SP	SP	SP	SP	SP	SP	SP
8	07:15-15:45	06:20-13:45	RD	RD	13:30-22:00	14:00-22:30	13:30-21:00
9	RD	12:00-20:30	12:00-20:30	12:00-20:30	12:00-20:30	12:00-20:30	RD
10	RD	14:00-22:30	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	14:45-23:55
11	RD	07:00-15:30	07:00-15:30	RD	RD	06:20-13:35	06:20-14:50
12	SP	SP	SP	SP	SP	SP	SP
13	RD	RD	14:00-22:30	13:30-22:00	14:00-22:30	14:00-22:30	14:30-22:00
14	RD	16:25-23:55	16:25-23:55	RD	RD	07:00-15:30	06:20-14:50
15	07:15-15:45	06:20-13:45	06:20-13:35	06:20-13:35	06:20-13:35	07:00-15:30	RD
16	RD	08:30-17:00	08:30-17:00	08:30-17:00	RD	RD	RD
17	10:30-19:00	16:25-23:55	RD	RD	06:20-14:20	06:20-14:20	06:20-14:50
18	SP	SP	SP	SP	SP	SP	SP
19	14:30-23:25	RD	06:20-13:45	06:20-13:45	06:20-13:45	06:20-13:45	08:00-16:00
20	08:30-16:30	06:20-13:45	RD	RD	13:30-22:00	16:25-23:55	13:30-22:00
21	13:30-22:00	13:30-22:00	RD	13:30-22:00	13:30-22:00	13:00-21:00	RD
22	RD	14:00-22:30	13:30-22:00	13:30-22:00	13:30-22:00	RD	RD
23	RD	RD	RD	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55
24	RD	RD	06:20-13:45	06:20-13:45	07:00-15:30	RD	06:20-13:45

Prepared 22 September 2026 · re-rendered 25 September 2026. The code **8eef9a13** in every footer is the design's fingerprint — eight characters computed from these 168 cells and nothing else. If the workspace shows the same code after you paste, you have exactly this design; if one cell differs, the code changes. Every figure in this document was computed by the Marylebone Roster app's Links modules (v24.22) from exactly these cells; import them and the workspace's Design checks, hard limits, fatigue factors and coverage cards will restate them.